

PROBE BUILD OF THE PROSE FILES

PROBE¹

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ABSTRACT

Could a nearby star be running a narrowband millimetre transmitter bright enough that ALMA has already recorded it? Almost nothing above 30 GHz has been searched, and no search there has reached stars this close. We have therefore searched the public ALMA archive at each star’s own position for an unresolved carrier whose frequency drifts: 402 spectral windows toward 60 stellar systems within 40 pc, with channels fine enough to follow a drift, covering 47.7 GHz of sky frequency in disconnected intervals from 90 to 873 GHz. Of the 56 threshold crossings, 19 fall on catalogued molecular transitions, including the β Pictoris disc, recovered blind; the rest are no more numerous than each field’s own control positions predict, though those positions are not a perfect null. Only 1 unattributed crossing is brighter at its star than at any control, and it is absent at the next observation of that frequency. No carrier both passed the selection and recurred in an independent observation. Sensitivity is measured by injecting carriers into the calibrated visibilities and recovering them through the whole selection: the median system’s best window reaches 1.77×10^{15} W, the power at which nine carriers in ten survive every stage — a recovery fraction, not a detection probability. Coarser windows add a supplementary limit on unresolved spectral excess. What limits such a search is line confusion and how often the archive revisits a star at one frequency, not sensitivity; and because ALMA chose these stars, the result describes them, not the solar neighbourhood.

Subject headings: extraterrestrial intelligence, radio lines: stars, submillimetre: stars, surveys, astrobiology

1. INTRODUCTION

This paper asks whether any nearby star is transmitting a narrow radio signal that ALMA happened to record while observing something else. The public archive holds thousands of hours of millimetre and submillimetre observations taken for quite different reasons, mostly studies of dust discs and planet formation, and every one of them also contains a spectrum of the star at the centre of the field. A transmitter would appear in such a spectrum as a spike confined to one or two channels. It would sit at the star’s position and nowhere else in the field, and reappear if the star were observed again. We searched the archive for exactly that. Nothing of the kind survived.

A technological civilisation may imprint the electromagnetic spectrum detectably whether or not it intends to, a *technosignature* (Sheikh 2020; Vidal et al. 2026), and millimetre wavelengths are an attractive place to look for one. Interstellar scattering and dispersion fall steeply with frequency, so a high-frequency carrier arrives less distorted; a given aperture delivers higher gain, and so more radiated power per watt supplied to the transmitter; and coherent millimetre emission is already engineered on Earth, for directional links, radar and beamed power. The band is correspondingly hard to search. The atmosphere transmits unevenly and adds phase noise, receivers are noisier than at centimetre wavelengths, and archived correlator channels are thousands of times coarser than a dedicated narrowband backend, so a carrier is diluted within one channel and its drift can smear it across several. These bands are also crowded with molecular lines, a dense background of real astro-

physical signals that a threshold search must separate from an artificial one.

Six decades of radio searches have examined many thousands of stars, yet sample only 6×10^{-18} of the space of transmitter frequency, bandwidth, power and duty cycle (Wright et al. 2018). Essentially all of that effort lies below ~ 30 GHz. Above it we are aware of one published search, in ALMA Band 3 (Mason et al. 2025), whose targets lie at 1.01 kpc and beyond; no search above ~ 30 GHz has yet reached a star within 40 pc at all. Few searches at any frequency reprocess existing interferometric archives, although those archives already hold a spectrum of every target they have ever pointed at (§1). That is the regime this paper enters.

The signal sought here is an *unresolved spectral carrier*. It is emission confined to a few channels, unresolved at the stellar position, and persistent enough to recur between epochs. Each qualifier is a real restriction. A carrier broader than the channel width would not be recovered, nor one that fills the synthesised beam, nor one that appears once and never again. Its frequency is allowed to drift, as it must for a transmitter carried on a rotating, orbiting body, and the drift searched is bounded by a line-of-sight acceleration rather than by a frequency (§1). The fine channels used are 15.3–1953 kHz wide, so “narrowband” here describes the assumed transmitter and never the resolving power of the data.

One limit on interpretation holds throughout. The targets were chosen by archival disc and planet-formation programmes, not for this search, and the sample inherits their preferences in distance and in spectral type (§1). Every statistical statement here is therefore about the ALMA archive-selected stellar sample and not about the solar neighbourhood.

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2. CONCLUSIONS

1. ALMA’s public archive opens a region of search space unexamined toward nearby stars. No search above about 30 GHz has reached a star within 40 pc, and the one previous Band 3 search targeted stars at 1.01 kpc and beyond. Of the 118.1 GHz of unique sky frequency searched here in either class, 20.6 GHz lies inside the frequency axis of the Wright et al. (2018) search haystack, which defines a parameter space and not coverage by any telescope.
2. The primary experiment is 402 fine-channel windows toward 60 stellar systems, covering 47.7 GHz of unique frequency in disconnected intervals, searched for an unresolved carrier whose frequency drifts. A secondary coarse-channel search for an unresolved spectral excess adds 70.4 GHz in 1249 windows, with no drift discrimination. Both come from one uniform reduction of 404 public execution blocks, 4.1 TB of raw data.
3. Sensitivity is a measured completeness, not a threshold. Carriers injected into the real calibrated visibilities are recovered through the unchanged selection, the spatial ranking included, and EIRP₉₀ is the power at which nine in ten come back. It is quoted per window: $3.45 P_{\text{trig}}$ where the control ring is noise, a bound where resolved disc emission raises it. The median system’s best window reaches 1.77×10^{15} W, with a spread across systems of 7.74×10^{13} – 5.79×10^{16} W.
4. All 56 threshold crossings are reported with their dispositions. Of these, 19 coincide with a catalogued molecular transition, almost all of them carbon monoxide. Of the remaining 37, 36 were tested for recurrence in the star’s own frame, at every later observation covering the event frequency: none is present again, the largest statistic at any predicted cell being 4.71 against $T_{\text{trig}} = 5$, and the weakest exclusion 5.0σ . The one crossing not tested had not retained its discovery spectrum, so there was no amplitude to carry into the epochs that do cover it. Exactly 1 of them outranks all 512 control positions, which is a prioritisation and not a confirmation.
5. What this survey could search and what it could confirm are different extents and must not be read as one statement. The search covered 60 systems, of which only 30, carrying 44 per cent of the fine-channel bandwidth counted window by window, were observed again at the same frequency more than a day apart, and 11 beyond a year. A transmitter is persistent here if it would be detectable above the quoted power in every independent observation covering its frequency.
6. Combining each star’s epochs in its own rest frame deepens the flux limit by a median factor 1.46. For Proxima Centauri the stacked limit, measured by injection into the stack, is $3.19\times$ deeper than that star’s own best single epoch and reaches 1.73×10^{13} W in 15.625 MHz channels — a Class B limit on an unresolved excess, carrying no drift discrimination — for an emitter steady in the stellar frame to 20.9 km s^{-1}

across 6.3 yr.

7. These results constrain persistent, unresolved carriers within the frequency, drift, duty-cycle and sensitivity domain actually searched, toward the stars ALMA happened to observe, and not the solar neighbourhood. The more transferable result is what such a search runs into. Thermal sensitivity does not bind it — ALMA supplies more than the experiment can use — but molecular-line confusion, a channelisation chosen by other observers for other purposes, and how often the archive happened to look twice.

AUTHOR CONTRIBUTIONS

G.J.W. conceived the survey, defined the target sample and the selection criteria, and wrote the manuscript. R.D. built and operated the archival retrieval, calibration and search pipeline, implemented the spatial control statistic and the injection–recovery campaigns, and produced the released catalogue and its regeneration scripts. Both authors designed the candidate vetting procedure, chose the masked species, carried out the statistical validation reported in §1 and Appendix 1, and agreed the dispositions of Table 1.

FUNDING

This work received no dedicated funding. It used public archival data and no observing time was awarded for it.

COMPETING INTERESTS

The authors declare no competing interests. VBRL Holdings is a privately held company through which R.D. conducts independent research, named here solely as his affiliation; it had no role in the design of the survey, the selection of targets, the analysis, the decision to publish or the content of this paper, and provided no funding for the work.

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This paper uses ALMA data from the public ALMA Science Archive. The 404 primary-census execution blocks searched span 2013 October 6 to 2025 June 11 across 63 proposal codes, which in the standard ADS/JAO.ALMA#*code* form are:

2012.1.00142.S,		
2012.1.00198.S,	2012.1.00268.S,	2012.1.00385.S,
2012.1.00437.S,	2012.1.00993.S,	2013.1.00359.S,
2013.1.00588.S,	2013.1.01147.S,	2013.1.01214.S,
2015.1.00307.S,	2015.1.00633.S,	2015.1.00783.S,
2015.1.01015.S,	2015.1.01260.S,	2016.1.00104.S,
2016.1.00358.S,	2016.1.00515.S,	2016.1.00637.S,
2016.1.00803.S,	2016.1.00817.S,	2016.1.00880.S,
2016.2.00015.S,	2016.2.00200.S,	2016.A.00013.S,
2017.1.00167.S,	2017.1.00200.S,	2017.1.00215.S,
2017.1.00561.S,	2017.1.00698.S,	2017.1.00704.S,
2017.1.00786.S,	2017.1.00986.S,	2017.1.01583.S,
2017.1.01644.S,	2018.1.00045.S,	2018.1.00072.S,
2018.1.00126.S,	2018.1.00461.S,	2018.1.00470.S,
2018.1.00947.S,	2018.1.01149.S,	2018.1.01545.S,
2018.1.01833.S,	2019.1.00189.S,	2019.1.01443.T,
2019.1.01517.S,	2019.1.01603.S,	2019.1.01681.S,
2019.2.00208.S,	2021.1.00485.S,	2021.1.01209.S,
2021.2.00097.S,	2022.1.00134.S,	2022.1.00338.L,
2022.1.00793.S,	2022.1.01163.S,	2023.1.00390.S,

2024.1.00165.S, 2024.1.00595.S, 2024.1.00722.S, 2024.1.01095.S, 2024.A.00040.S. Per-block epochs are in the machine-readable release. The archive meta-data exposes no proposal code for 2 of the 404 blocks (A002_X11d9ce7_X5257, A002_Xbf792a_X26ec), which are therefore not represented in that list. ALMA is a partnership of ESO (representing its member states), NSF (USA) and NINS (Japan), together with NRC (Canada), NSTC and ASIAA (Taiwan), and KASI (Republic of Korea), in cooperation with the Republic of Chile. The Joint ALMA Observatory is operated by ESO, AUI/NRAO and NAOJ. This work has made use of data from the European Space Agency (ESA) mission *Gaia* (Gaia Collaboration et al. 2023), processed by the *Gaia* Data Processing and Analysis Consortium (DPAC). This research has made use of the SIMBAD database (CDS, Strasbourg, France).

DATA AVAILABILITY

The reference of record is a single permanent Zenodo deposit, cited by its DOI. It carries every number in this paper and the code that produced them, and contains: the 89-star target catalogue with its selection function; one row for each of the 1651 searched windows, carrying the ALMA project code and execution-block identifier, frequency limits, native channel width, trial-drift grid, trigger statistic at the stellar position, control statistics of the 512-position ring and the window’s EIRP₉₀; all 56 threshold crossings with their stellar-frame frequencies, line classifications, recurrence measurements and dispositions; the injection–recovery records; the block-

by-block archival verdicts and the ALMA delivery record for every block in scope that could not be searched; and the full pipeline with the commit that produced the deposited products. What has been demonstrated rather than claimed is narrower, and it is a measured number in both cases: 120 generated products regenerate byte-identically from the frozen inputs, and 41 headline quantities are re-derived from the released catalogue alone, without the pipeline.

The choices that could manufacture a candidate were fixed in advance, and two of them can be checked without the deposit: the calibration hold-out rule of Appendix 1 and the block list its first guard depends on were committed to the project repository as c75069040eab52ec1336376a4dae62114bb281a3 on 2026 September 14 at 07:10 UTC, 3 days before the first reserved block was searched. The frozen search statistic and the frozen line list are deposited with the data release, which is the identifier for both. The repository at <https://github.com/Tilanthi/SETI> is a working mirror and may move on.

SOFTWARE

The pipeline depends on CASA 6.7.6 (casatools/casatasks 6.7.6-14; CASA Team et al. 2022), Python 3 with astropy (Astropy Collaboration et al. 2022), numpy (Harris et al. 2020) and scipy (Virtanen et al. 2020), and the ALMA Science Archive and SIMBAD TAP interfaces through astroquery (Ginsburg et al. 2019); all are free and open-source, with exact versions recorded in the frozen release.

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